Year 13 Further Maths - Roots of polynomials

Similar questions - Complex roots and conjugate pairs

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning. You are given that $x = 3 + 2\mathrm{i}$ is a root of $x^3 - 7x^2 + 19x - 13 = 0$. Solve the equation. **[4]**

2 In this question you must show detailed reasoning. The equation $z^4 + 8z^3 + 29z^2 + 52z + 42 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

(a) Show that the quartic equation with roots $\alpha + 2$, $\beta + 2$, $\gamma + 2$, $\delta + 2$ is $w^4 + 5w^2 + 6 = 0$. **[3]**

(b) Hence determine the exact roots of $z^4 + 8z^3 + 29z^2 + 52z + 42 = 0$. **[4]**

3 The cubic equation $2x^3 + px^2 + qx - 3 = 0$, where p and q are real, has a complex root $\alpha = 1 + \mathrm{i}\sqrt{2}$.

(a) Write down a second complex root β. **[1]**

(b) Determine the third root γ. **[2]**

(c) Find p and q. **[2]**

(d) Show that for any integer n, $\alpha^n + \beta^n + \gamma^n = 2 \times 3^{n/2} \cos n\theta + \dfrac{1}{2^n}$ where $\tan\theta = \sqrt{2}$. **[4]**

Question 1
Coefficients are real, so $3-2\mathrm{i}$ is also a root. $(x-3-2\mathrm{i})(x-3+2\mathrm{i}) = x^2-6x+13$. Dividing, $x^3-7x^2+19x-13 = (x^2-6x+13)(x-1)$. Roots: $x = 3\pm 2\mathrm{i}$ and $x = 1$

Question 2
(a) Let $w = z+2$, so $z = w-2$: $(w-2)^4+8(w-2)^3+29(w-2)^2+52(w-2)+42$. Expanding: $w^4-8w^3+24w^2-32w+16+8w^3-48w^2+96w-64+29w^2-116w+116+52w-104+42 = w^4+5w^2+6$, so $w^4+5w^2+6=0$
(b) $(w^2+2)(w^2+3)=0$, so $w = \pm\mathrm{i}\sqrt{2}$ or $\pm\mathrm{i}\sqrt{3}$. Then $z = w-2$: $z = -2\pm\mathrm{i}\sqrt{2}$ and $z = -2\pm\mathrm{i}\sqrt{3}$

Question 3
(a) $\beta = 1-\mathrm{i}\sqrt{2}$
(b) Product of roots $= -\dfrac{-3}{2} = \frac{3}{2}$. $\alpha\beta = 1+2 = 3$, so $\gamma = \frac{1}{2}$
(c) Sum of roots $= 2+\frac{1}{2} = \frac{5}{2} = -\dfrac{p}{2}$, so $p=-5$. $\alpha\beta+\gamma(\alpha+\beta) = 3+\frac{1}{2}\times 2 = 4 = \dfrac{q}{2}$, so $q=8$
(d) $|\alpha| = \sqrt{3}$ and $\arg\alpha = \theta$ with $\tan\theta = \sqrt{2}$, so $\alpha = \sqrt{3}\mathrm{e}^{\mathrm{i}\theta}$ and $\beta = \sqrt{3}\mathrm{e}^{-\mathrm{i}\theta}$. Then $\alpha^n+\beta^n = 3^{n/2}(\mathrm{e}^{\mathrm{i}n\theta}+\mathrm{e}^{-\mathrm{i}n\theta}) = 2\times 3^{n/2}\cos n\theta$ (de Moivre) and $\gamma^n = \dfrac{1}{2^n}$. Adding gives the result